

EG800Q Series

Reference Design

LTE Standard Products

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About the Document

Revision History

Version	Date	Author	Description
-	2022-07-14	Chan Chen	Creation of the document
1.0	2023-05-30	Maffie Zhang	First official release
1.1	2024-08-28	Fanny Chen/ Lem Jin	- Added the applicable module EG800Q-NA. - Added a note about Wi-Fi scan antenna (Sheet 1). - Updated the current that is provided to the module from 2.0 A to 1.5 A (Sheets 1 and 5). - Updated the I/O direction of TXD_MCU and RXD_MCU (Sheets 4 and 6). - Updated the antenna interface design and added related notes (Sheet 7).
1.2	2025-11-27	Fanny Chen	- Added the applicable module EG800Q-IN. - Updated the width of VBAT trace to 1.5 mm (Sheet 5).

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1 Reference Design

1.1. Introduction

This document provides the reference design for Quectel EG800Q series module. The reference design mainly includes block diagrams, power supply design, UART and USIM interface design, antenna interface design, audio codec design and other designs.

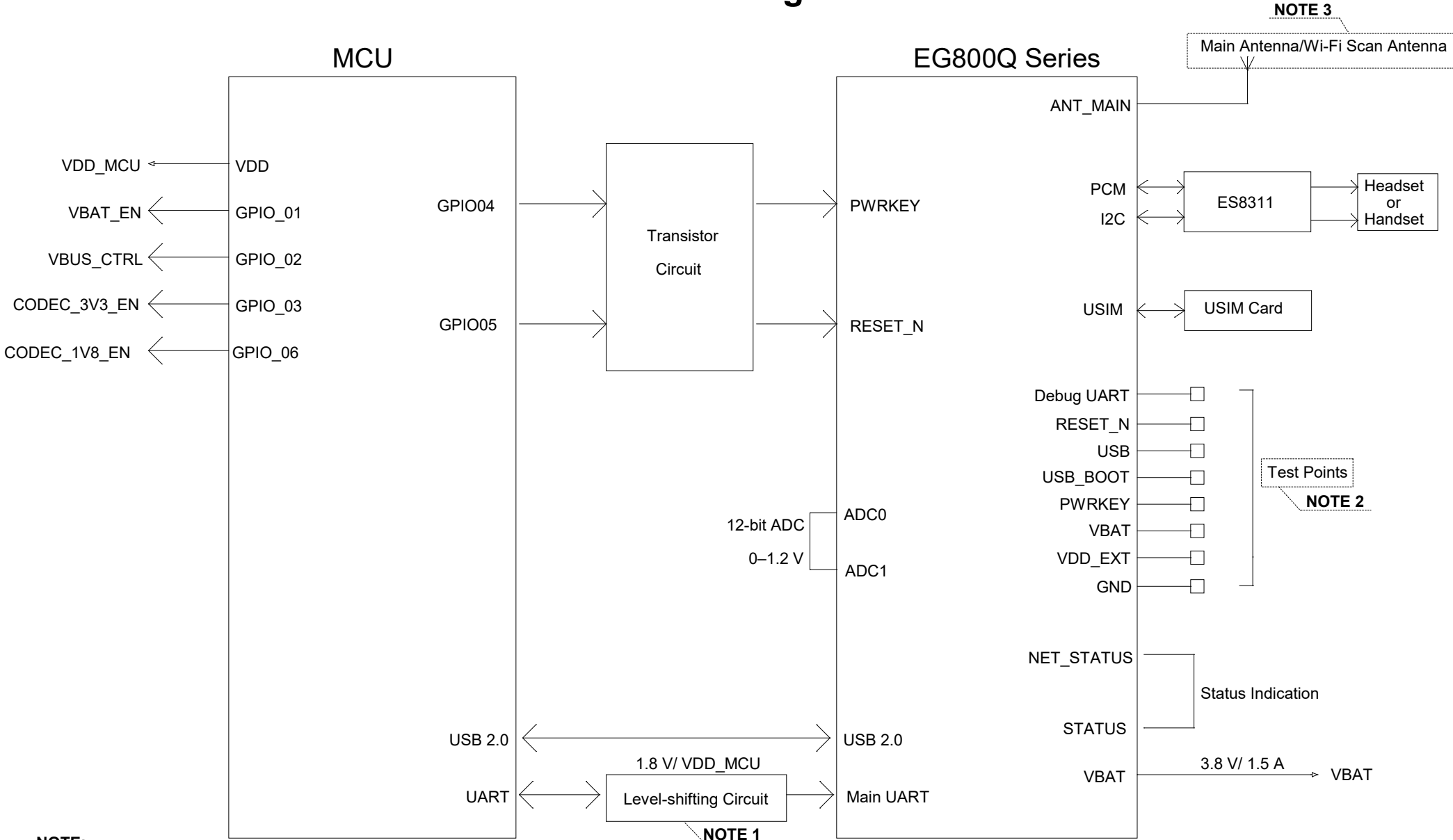
1.2. Schematics

The schematics illustrated in the following pages are provided for reference only.

NOTE

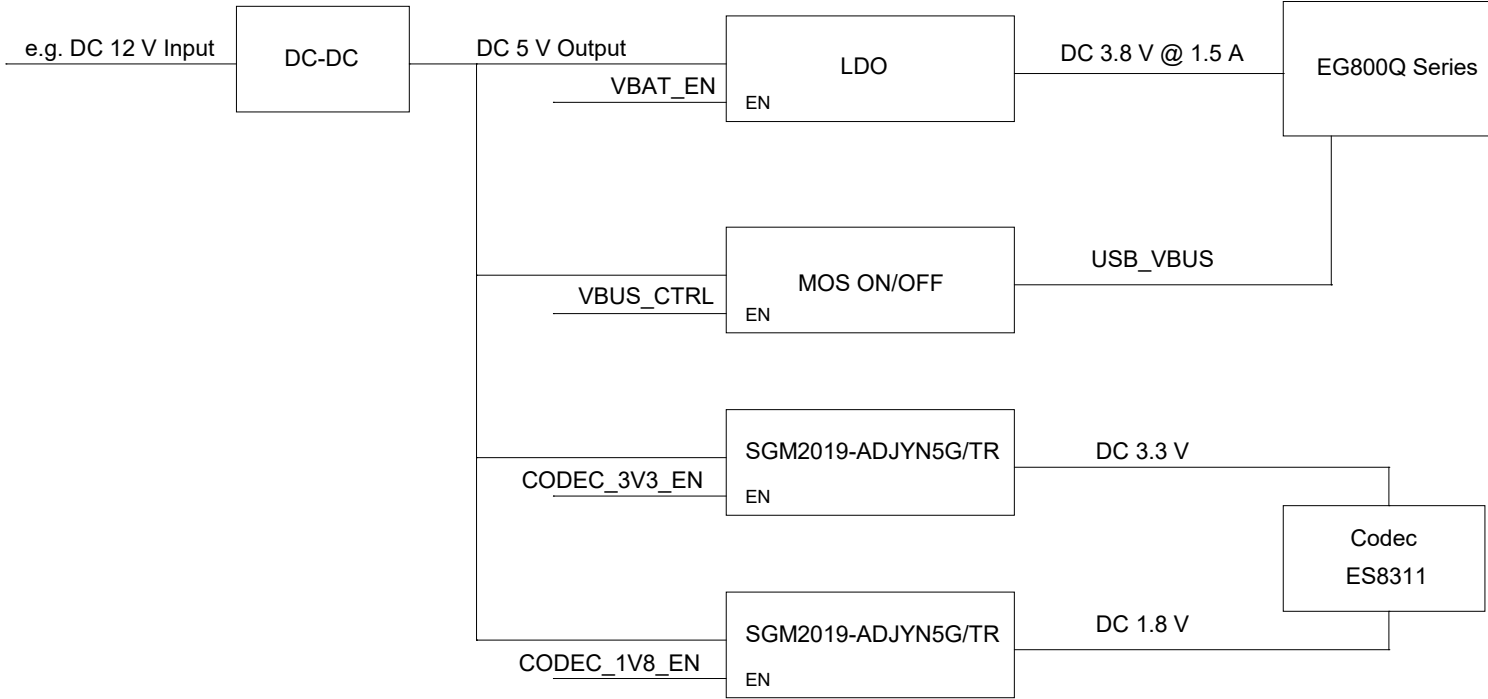
It is required to confirm the applicability and price from the supplier about the IC involved in the reference design.

Block Diagram



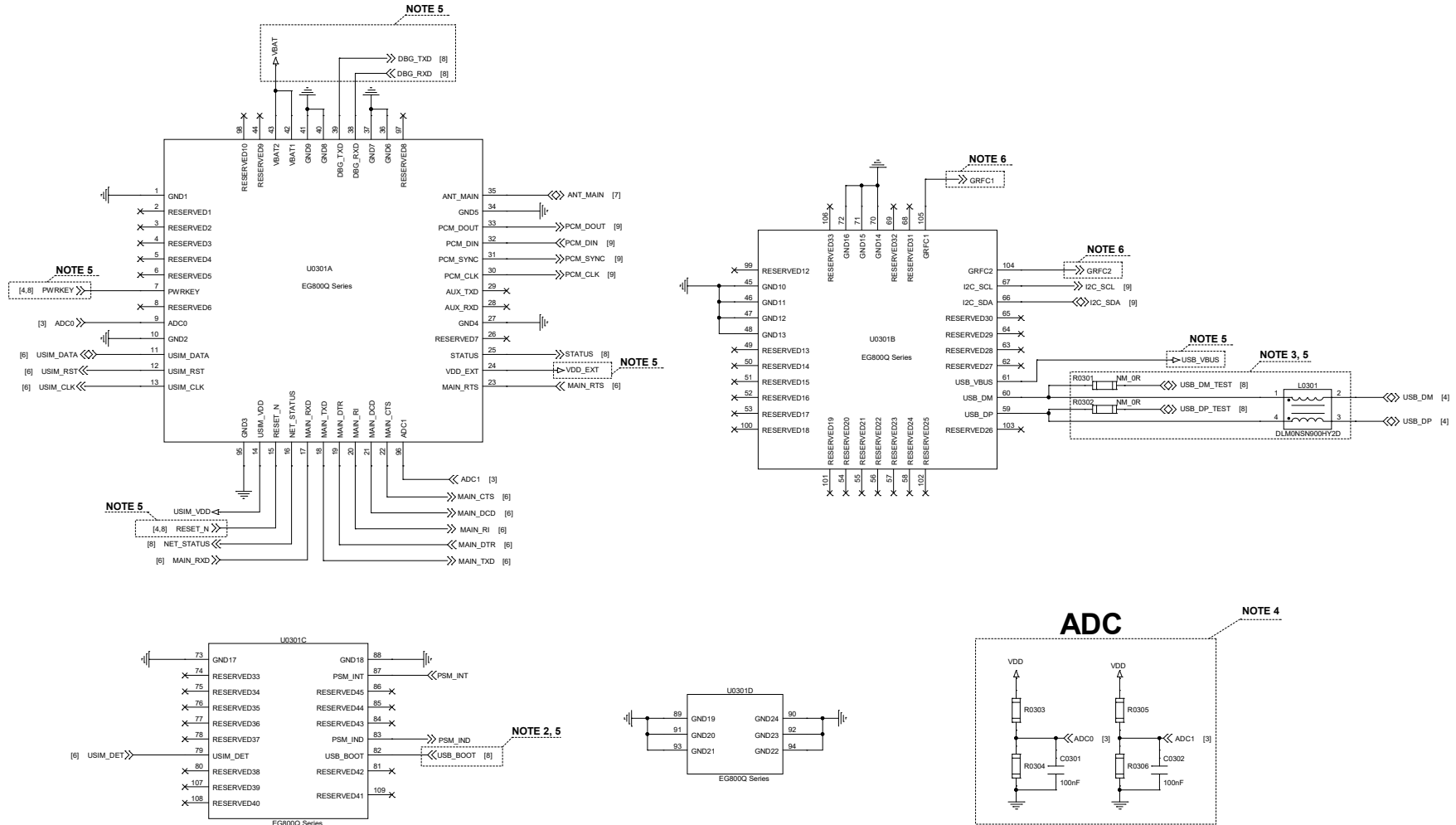
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Power System Block Diagram



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Module Interfaces

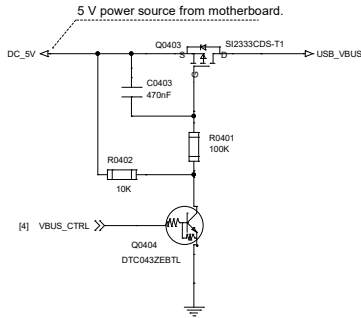
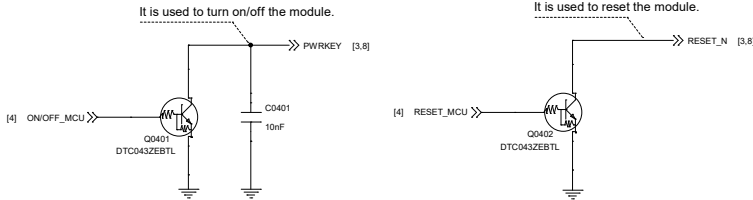
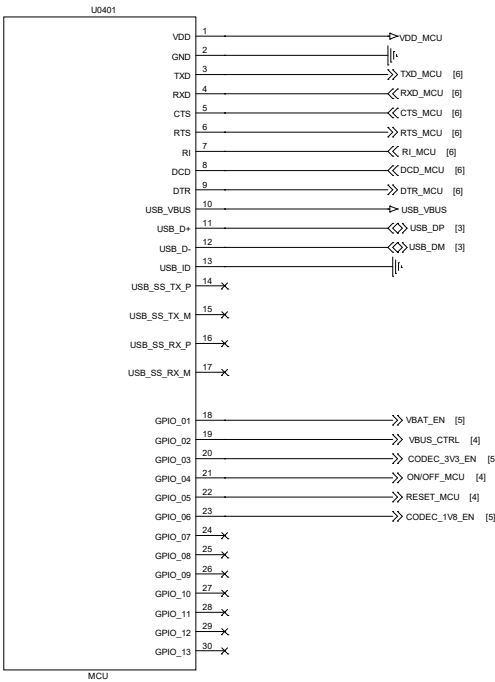


NOTE:

1. All GND pins should be connected to ground, and keep unused and RESERVED pins unconnected.
2. If the module does not need to enter forced download mode, do not drive USB_BOOT pin low before turning on the module.
3. It is recommended to add a common mode choke L0301 in series between the module and your MCU to suppress EMI. Additionally, test points must be reserved over USB_DP and USB_DM for firmware upgrades, and it is recommended to minimize extra trace stubs.
Place L0301 and two resistors, R0301 and R0302, close to the module to ensure USB signal integrity.
4. The input voltage range of ADC0 and ADC1 interfaces is 0–1.2 V. When the collecting voltage is higher than 1.2 V, it is recommended to use the resistor divider circuit for ADC application.
The divider resistor accuracy should be less than 1 %, and the resistance should not exceed 100 kΩ. It is recommended to reserve a 100 nF capacitor for the design.
5. Test points of USB_VBUS, USB_DP/DM, USB_BOOT, DBG_TXD/RXD must be reserved.
It is recommended to reserve test points for PWRKEY, VBAT and VDD_EXT.
It is recommended to reserve a test point for RESET_N if unused.
6. If you need GRFC function, please contact Quectel Technical Support for the related reference design.

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MCU Interfaces



NOTE:

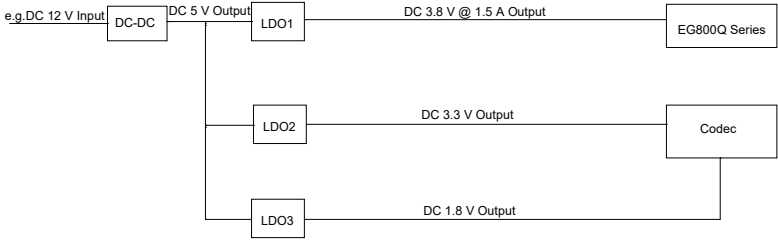
- If the power domain of your MCU (U0401) is also 1.8 V, the level-shifting circuit is not necessary as it matches the 1.8 V power domain of the module's GPIO interfaces.
- The module only support USB slave mode and supports full-speed and high-speed modes of USB 2.0. To communicate with the USB interface, MCU needs to support USB host mode or OTG function.
The USB_VBUS pin of the module should be powered by an external power system for USB detection. Use VBUS_CTRL to control the on/off state of the USB_VBUS ower supply.
- The reset function of the module requires the PWRKEY and RESET_N pins to work together. For more details, see the hardware design document of the module.
- It is recommended to select the MCU's default low-level GPIO pins as the control pins for the module's PWRKEY and RESET_N pins. Ensure that the load capacitance on PWRKEY and RESET_N pins does not exceed 10 nF.

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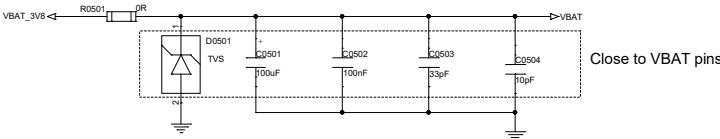
Power Supply Design

DC-DC Application

When the input voltage is above 7.0 V, use a DC-DC converter to convert the high input voltage to 5.0 V, and then use LDOs to convert it to 3.8 V, 3.3 V and 1.8 V to power the module and Codec.



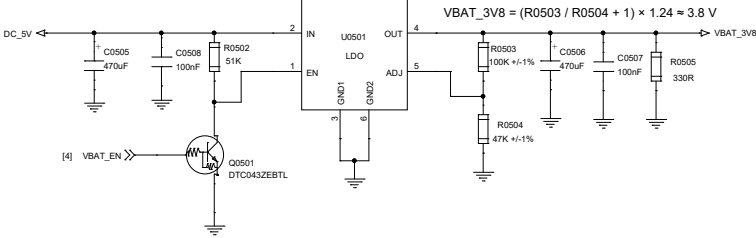
VBAT Design



- NOTE:**
1. It is recommended to provide 1.5 A current to the module.
 2. The width of VBAT trace should be at least 1.5 mm.
 3. The recommended operating voltage of VBAT is 3.3–4.3 V, and the typical operating voltage is 3.8 V.

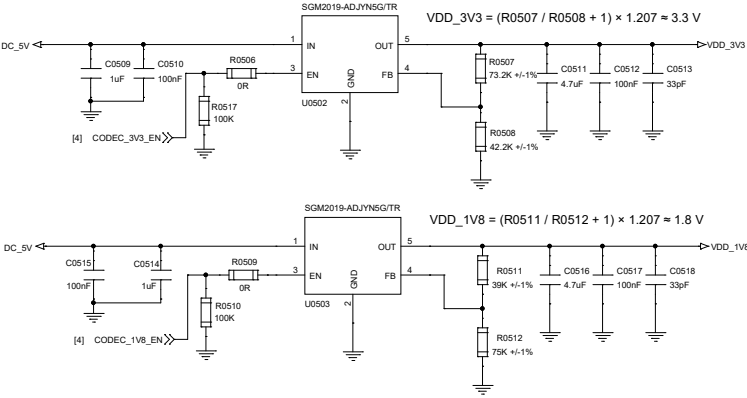
LDO Application

When the input voltage is below 7.0 V, use an LDO to convert the input voltage to 3.8 V.



- NOTE:**
1. The recommended load current should exceed 10 mA.
 2. You can adjust related device parameters according to the actual design.

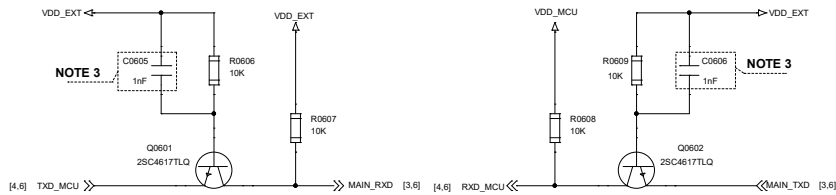
Power Supply for PCM Codec



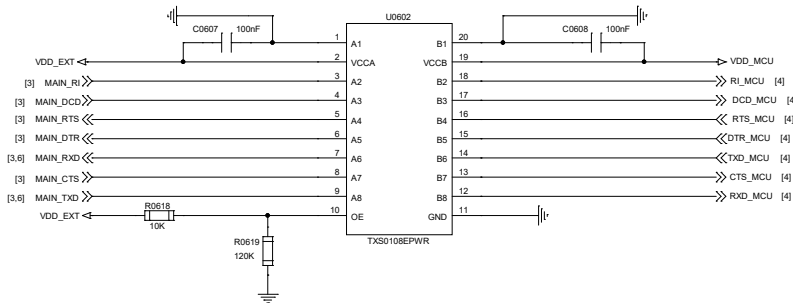
- NOTE:**
1. CODEC_1V8_EN is used to turn on/off the VDD_1V8 power supply.
CODEC_3V3_EN is used to turn on/off the VDD_3V3 power supply.
 2. For ES8311 codec, the power-up/down interval between 1.8 V and 3.3 V power supplies should not exceed 10 ms.

UART and USIM Interface Design

UART Level-shifting Circuit - Transistor Solution



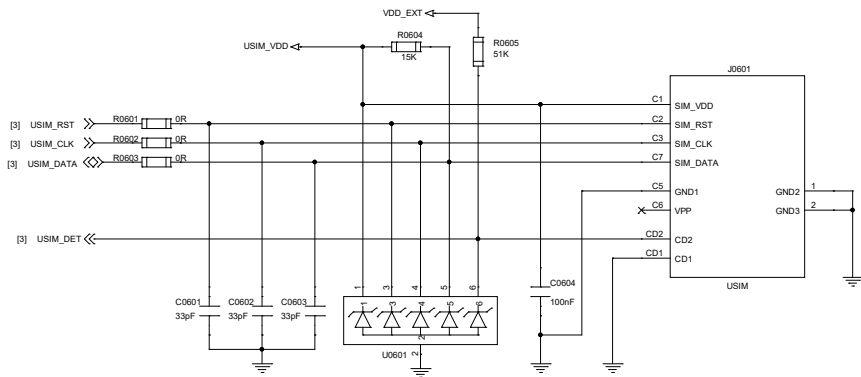
UART Level-shifting Circuit - IC Solution



NOTE:

- There are two level-shifting circuits: transistor solution and IC solution, and it is recommended to select the latter one.
- The power supply of TXS0108EPWR's VCCA should not exceed that of VCCB. For more information, see the datasheet of TXS0108EPWR.
- The transistor solution is not suitable for applications with baud rates exceeding 460 kbps.
The 1 nF capacitors (C0605 and C0606) can improve the signal quality.
- The transistor solution of MAIN_RTS is similar to that of MAIN_RXD.
The transistor solutions of MAIN_CTS, MAIN_RI and MAIN_DCD are similar to that of MAIN_TXD.
- If the level of the external host is 1.8 V and the module's MAIN_RXD and MAIN_TXD are directly connected to the host's TXD and RXD, the MAIN_TXD of the module should be connected to a 10 kΩ resistor and pulled up to 1.8 V to prevent the host from receiving error messages when the module is in sleep mode.
- The level-shifting circuits take the main UART as an example. The circuits of the debug UART and the auxiliary UART are connected in the same way as the main UART.

USIM Interface Design



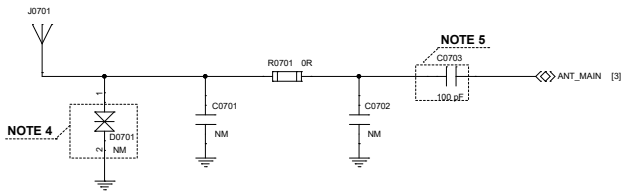
NOTE:

- U0601 is recommended to be used to offer good ESD protection, and the parasitic capacitance should be less than 15 pF.
- The pull-up resistor R0604 can improve anti-jamming capability, and should be placed close to the USIM card connector.
- C0604's capacitance should be less than 1 μF and it should be placed close to the USIM card connector.
- R0601–R0603 are used for debugging, and C0601–C0603 are used for filtering out RF interference.

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Antenna Interface Design

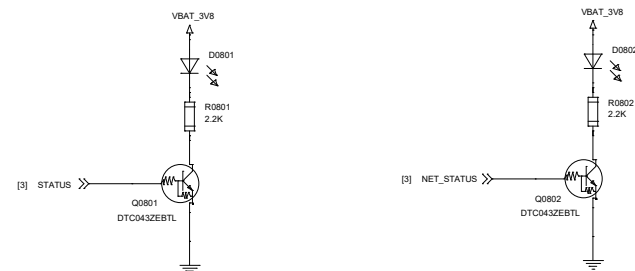


- NOTE:**
1. The single-ended impedance of the RF antenna is 50 Ω .
 2. It is highly recommended to reserve a dual L-type circuit for main antenna for future debugging.
 3. The impedance of RF signal traces must be controlled as 50 Ω when routing.
 4. It is recommended to reserve an ESD protection component for the cellular antenna interface to effectively prevent static electricity and the junction capacitance should not exceed 0.05 pF.
 5. Notes on C0703:
 - 1) If there is DC power at the antenna port, place capacitor on C0703 to prevent short circuit to ground. The capacitance value is recommended to be 100 pF, which can be adjusted according to the debugging results.
 - 2) If there is no DC power in the peripheral design:
 - a) Do not reserve C0703.
 - b) If C0703 has already been reserved, it should be mounted with component, and it is recommended to use 0 Ω resistor. You can also match the component according to the debugging results.

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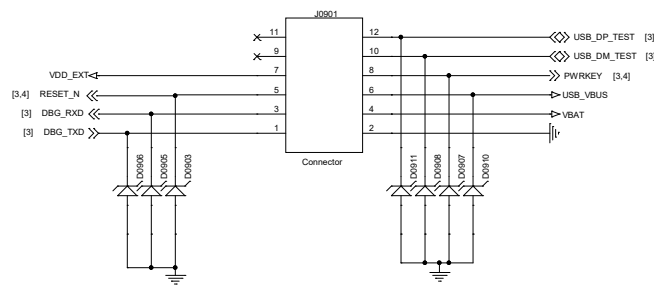
Other Designs

Indicators



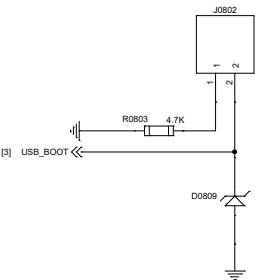
NOTE:
To minimize the module's power consumption during sleep mode of your device, replace the power supplies of NET_STATUS and STATUS indicators with external controllable sources and turn off the indicators when the module is in sleep mode.

Reserved Test Points



- NOTE:**
- Test points for both USB and debug UART interfaces are reserved for outputting partial logs.
 - Test points for USB interface must be reserved for firmware upgrade.
 - The stray capacitance of the ESD protection components on USB data traces should be less than 2 pF.
 - The debug UART interface supports a 1.8 V power domain. If your application operates at 3.3 V, use a voltage-level translator.

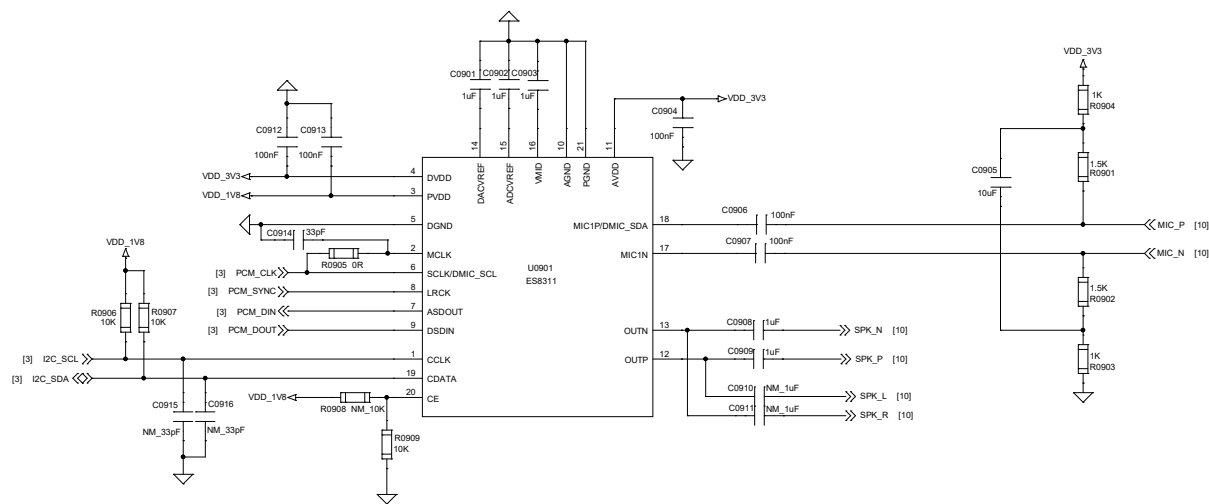
Forced Download Interface



- NOTE:**
- A test point of USB_BOOT must be reserved.
 - Before turning on the module, pull USB_BOOT down to GND to activate the forced download mode.
In this mode, the module supports firmware upgrade over USB 2.0 interface.

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Audio Codec Design (ES8311)



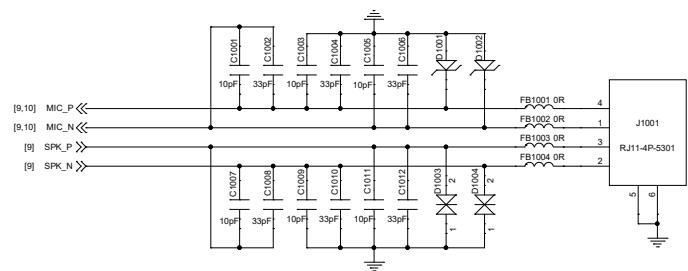
NOTE:

- 1. ES8311 power-up/down sequence: AVDD, PVDD and DVDD are powered up and down at the same time, or with the interval within 10 ms.
- 2. When the sampling frequency is 8 kHz and the clock frequency is 512 kHz, DVDD must be connected to 1.8 V.
- 3. The bias voltage needs to be stable and the ripple should be as small as possible.
- 4. AGND and DGND are connected through a 0 Ω (0805 package) resistor, please refer to the Audio Codec Design page for design.
- 5. For more details, see the datasheet of ES8311.

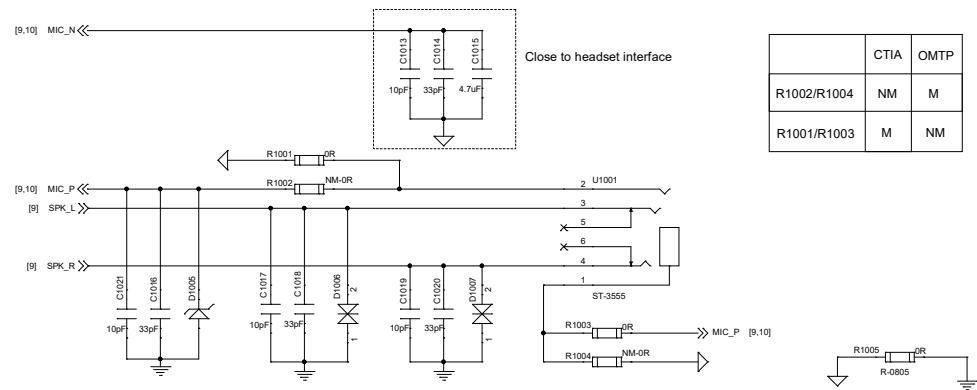
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Audio Codec Design (Analog Audio Interface)

Handset Application



Headset Application



- NOTE:**
1. The analog output signals can drive handset and headset. For larger power loads such as loudspeaker, an audio power amplifier should be added in the design.
 2. In handset applications, route the MIC and SPK signal traces as differential pairs respectively.
 3. In headset applications, route the MIC signal traces as a differential pair.
 4. All MIC and SPK signal traces should be surrounded with ground on that layer and with ground planes above and below, and should be far away from noise such as clock and DC-DC signals.
 5. AGND and DGND are connected through a 0 Ω (0805 package) resistor.

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